



Evaluation of social media information among college students: An information literacy approach using CCOW

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ABSTRACT

Social media platforms have become increasingly important as sources of information in higher education. However, assessing the quality of information on these platforms posed a challenge due to wide variability in reliability. This study aimed to explore the impact of the CCOW (Credentials, Claims, Objectives, Worldview) approach on college students' evaluation of social media information. Participants were provided with a CCOW information literacy guide, which served as a manual for analyzing social media data. Prior to the intervention, students completed a pre-test assignment to assess their proficiency in evaluating social media material without the CCOW Guide. Subsequently, a post-test assignment was administered after the intervention, utilizing the CCOW guide to evaluate material from social media. The primary goal of the study was to determine the effect of the CCOW approach on students' evaluation skills. The findings highlighted significant differences in the evaluation of social media information following the implementation of the CCOW information literacy approach.

Introduction

As technology has advanced, various channels have emerged to provide access to an increasing amount of information, users can now access information not only through traditional channels such as printed sources but also sources in various formats via the internet (Kim et al., 2021) social media has emerged as one such avenue; it has indeed evolved beyond its original purpose of facilitating socialization and has become a powerful tool for information access and dissemination (Kim & Sin, 2015). Social media whose content is contributed by end-users has become increasingly popular as information sources, its enhanced learning systems allow students to participate in educational online communities by creating, editing, and sharing content online, communicating and exchanging opinions, connecting, establishing social networks, and creating communities for different needs (Silius et al., 2011) social media usage is increasing, and its diversity and mobility are contributing to this growth (Nielsen, 2012) it has emerged as the main information sources and avenues for communication, enabling users to connect and share information in a pervasive and convenient way (Hong, 2015).

Acquisition and evaluation of social media information

Undoubtedly, the rise of social media platforms like Facebook, Twitter, Instagram, and TikTok has fundamentally transformed how students access and gather information (Manca & Ranieri, 2016). These platforms provide a wealth of diverse topics, making information readily available to students at their fingertips. Consequently, this prompted a substantial shift in information acquisition from traditional to social media, with students increasingly depending on these platforms for their information needs (Harper, 2010). The transition from traditional media to social media can be attributed to the transformative power of the digital landscape, with social media; users can effortlessly access an ever-expanding wealth of information through user-friendly and cost-effective formats (Nielsen et al., 2016) this effortless access to a wide range of information on social media has resulted in a substantial number of college students relying on these platforms for various types of information, including academic and everyday topics (Head & Eisenberg, 2010; Jones & Fox, 2009; Brenner, 2012, as cited in Kim et al., 2014). Further, Kim et al. (2011) stated that students frequently utilize social media as a source of information for their academic work due to its accessibility and usability, and because they are inexpensive and practical tools for acquiring pertinent information (Ansari & Khan, 2020). This trend is developing as a result of various factors, including

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easy access, mobility, visual sharing and feedback, and the fact that social media can also serve as a platform for collaboration and discussion, allowing students to engage with others and share their ideas and perspectives (Greenhow & Askari, 2017; Kirschner & Karpinski, 2010) this kind of exchange of ideas and information can be a valuable learning experience, as students can learn from each other's unique perspectives and knowledge. Notably, modern students tend to be more selective in the information they consume, focusing on topics that align with their interests and needs, additionally, they value their ability to contribute to the content they consume and share their opinions on the matter (Ali & Yousaf, 2017), this preference has led students to favor social media as a source of information as it offers a diverse range of content that can be tailored to individual interests and preferences, and students can actively engage with this content by commenting, sharing, and creating their posts. This active participation in the online discourse allows students to contribute their opinions and perspectives, which can further shape the content they consume. As a result, social media offers a personalized and interactive approach to learning and knowledge acquisition. Moreover, students gain exposure to a variety of perspectives and viewpoints on various issues, often encountering unique insights from influencers and subject-matter experts in their field (Kirschner & Karpinski, 2010). However, over-reliance on social media for information acquisition comes with drawbacks, a notable concern is the risk of distraction due to constant notifications and updates, leading to decreased focus, procrastination, and reduced productivity, which can negatively affect academic performance (Junco, 2012).

It is a reasonable inference that a large proportion of youths rely on their smartphones to obtain information. Many assume that because young people are fluent in social media they are equally savvy about what they find there. However, it remains unclear if they examine each piece of information with the same level of scrutiny as they would during a formal academic research project (Johnson & Ewbank, 2018). The growing trend of students relying on social media platforms for information has become a major concern for educators and library professionals, primarily due to the trustworthiness of information such as the inconsistent quality and reliability of the information that is accessible through these channels. With so much content being shared on social media platforms, it can be difficult to discern what is accurate and what is not. The apprehension of educators and librarians regarding the use of social media as a source of information is compounded by the realization that students' ability to evaluate information has historically been less than satisfactory, even before the advent of social media (Metzger et al., 2003; Tomauiuolo, 2005). This issue is further exacerbated by the tendency of students to neglect the critical process of evaluating the accuracy and reliability of information obtained from online sources, as highlighted in research by Metzger et al. (2003). When the student gets the same inaccurate information from several different sources, unsuspecting students may misjudge the information as credible based on mere repetition (Begg et al., 1992; Channouf & Rouibah, 1995; Dechène et al., 2010). The brevity of social media messages presents another significant challenge in assessing their credibility, as their succinct nature can limit the amount of information available for critical evaluation. In addition, the rapid spread of rumors and misinformation across various social networks and channels can exacerbate the difficulty of evaluating the accuracy and reliability of information obtained from these sources (Nahon & Hemsley, 2013).

With the rise of new media, students now have more access to information without having to rely on traditional sources and since social media information undergoes little scrutiny, the quality of the information varies widely (Kim et al., 2014). This means that college students need to be more responsible for evaluating the credibility of the information they find on social media because there are fewer professional gatekeepers to check the contents. In this digital media landscape, information consumers act as their gatekeepers (Kovach & Rosenstiel, 1999), constantly monitoring the gates through which information passes and deciding what information is relevant and useful to share

with others. This is known as "gatewatching" (Bruns, 2008). Gate-watchers do not create new information themselves but rather share and add to existing information that they find useful or interesting. This is a distinctive feature of social media where users co-create content. However, with an enormous amount of information available online, it is challenging for users to organize the information and assess the credibility of its sources. Since there is a lack of consistency in the quality of information, users have to continually monitor and evaluate the credibility of the information they consume (Westerman et al., 2014).

CCOW: what and why?

The existing body of information literacy literature provides a plethora of techniques for evaluating information, with checklist approaches being notably prevalent and widely embraced methods. As the exploration of effective source evaluation continues, the prevalence and importance of checklist approaches persist as prominent features within scholarly discussions and pedagogical practices reflecting their enduring importance in the field of information evaluation. Among the various checklist methods, the CARS Checklist (Credibility, Accuracy, Reasonableness, Support) introduced by Harris in 2018, and the RADAR approach (Relevance, Authority, Date, Appearance, Reason for Writing) suggested by Mandalios in 2013 are noteworthy examples. However, the CRAAP method (Currency, Relevance, Authority, Accuracy, Purpose) formulated by Blakeslee in 2004 has emerged as the most widely employed and favored approach of them all. Its comprehensive and systematic framework has solidified its status as the go-to method for evaluating information across diverse circumstances. The implementation of the CRAAP test is extensively taught by experts in library and information literacy fields to assess online resources, research papers, and other complex academic projects. Its widespread use has gained significant momentum across diverse educational institutions and campuses (Doyle & Hammond, 2006; Blakeslee, 2004; Dinkelman, 2010). However, the CRAAP checklist has faced substantial criticism for being insufficient and inadequate in addressing the complexities of the present-day information landscape (Caulfield, 2016; Crum, 2017). This pertains to the fact that the CRAAP test was initially developed rooted in an era during the shift from Web 1.0 to Web 2.0; during this period the world underwent a significant transformation in the digital landscape, most online content was primarily intended for consumption rather than active interaction, modification, sharing, and when online information is treated as equivalent to printed texts and emphasizing careful reading for evaluation (Valenza, 2020). Bull (2021) point out that although the CRAAP test is designed to lessen cognitive overload, it might have the opposite effect, exacerbating cognitive burden. Consequently, students may end up making sub-optimal decisions concerning the credibility of sources, especially in intricate and interconnected online environments. Seeber (2017) further added that the CRAAP test approach relies on simplistic and binary criteria for evaluating sources. Instead of encouraging deep critical thinking and nuanced analysis, it tends to categorize sources into rigid and dichotomous classifications. The CRAAP test may not fully capture the complexities of information evaluation in real-world scenarios where truth, credibility, and reliability are often multifaceted and require more comprehensive assessment. It's approaches are insufficient when confronted with the vast expanse of the web, teeming with hoaxes, deep fakes, cloaked sites, source hackers, and scammers. The complexity and diversity of online content render such guidelines ineffective in addressing the challenges posed by the ever-evolving landscape of misinformation and deceptive sources (Wineburg et al., 2020). In this context, the article describes a new approach to source evaluation that builds upon the existing CRAAP (Currency, Relevance, Authority, Accuracy, Purpose) test, the new approach, called CCOW (Credentials, Claims, Objectives, and Worldview) was developed by Tardiff (2022), it adds a new criterion and redefines or removes some of the existing criteria to better reflect the process of active investigation and metacognition.

The CCOW approach presents each criterion as an object of active investigation and provides students with a set of tasks to perform during the investigative process. For example, the Credentials criterion in CCOW, as well as the Authority criterion in CRAAP, shares a common focus on evaluating the expertise and qualifications of the author or source of information. Both criteria aim to assess whether the source is credible and knowledgeable in the subject matter. However, the CCOW's Credentials criterion expands beyond solely considering the authority of the author or source; it asks students to investigate the qualifications of the author or organization and to look for any indications of unreliability. It not only delves into the qualifications of the author but also considers additional factors that contribute to the source's reliability and trustworthiness. Unlike the Authority criterion in CRAAP, which mainly focuses on the author's credentials and institutional affiliation, without necessarily exploring other aspects that could impact the reliability of the information presented, the Credentials criterion in CCOW takes a more comprehensive approach to evaluate source reliability by considering multiple factors that impact trustworthiness. Credentials criterion provides a more nuanced and robust assessment of information sources in the complex online world. The Claims criterion tasks students with finding multiple sources and putting them in conversation to evaluate the claims made in the source. While the CRAAP test lacks an exact equivalent for the Claims criterion present in CCOW, it can be related to the aspect of the Accuracy criteria in CRAAP. Accuracy in CRAAP pertains to the factual correctness of the information, which aligns with evaluating the accuracy of claims in CCOW. While both criteria aim to ensure the reliability of the information, the Claim criterion in CCOW goes beyond simply verifying the factual correctness of the content; by engaging in investigation for additional sources and comparing them to corroborate or challenge the claims made in a particular source. The Objectives criterion prompts students to consider the purpose of the information and to determine whether it was created to inform, convince, or sell.

The two criteria removed from the CRAAP test were Relevance and Currency. Relevance, which asks students if the information fits their current needs, was deemed useful for library acquisitions but not as directly related to source evaluation. Currency, which asks students when the information was published or posted, was folded into the Claims criterion, as currency's impact on a source's reliability is contextual and can distract students from more consistently useful evaluation criteria.

The CCOW approach also adds a new criterion, Worldview, which gives metacognition an explicit place in the evaluation process and encourages students to consider their internal response to the information as they evaluate it. Worldview is defined as "the overall perspective from which one sees and interprets the world" (*American Heritage Dictionary of the English Language*, 2011) and provides a more impartial and intricate method of examining the notion of bias. By framing the concept of bias as a natural and expected lens through which individuals view reality, rather than a negative characteristic; the CCOW approach aims to prompt students to a more careful consideration of their own biases and those of others.

The concept of considering the information object as the agent that is acting on the individual is an important one in today's information landscape, especially with the prevalence of algorithmic personalization and the efforts of various interest groups to shape and manipulate information (Tardiff, 2022). This means that individuals are often being exposed to information that has been carefully crafted and targeted toward them, which can make it difficult to discern what is true and what is not. Bull (2021) argue that the traditional reactive approach of SIFT and CRAAP, where the individual is responsible for evaluating the information they encounter, is no longer sufficient. Instead, they suggest that we need to adopt a more proactive approach that takes into account how information is being actively directed toward us. This means looking not only at the characteristics of the information itself but also at the broader context in which it is being presented and how it is being

disseminated.

In light of the web's abundance of hoaxes, deep fakes, cloaked sites, source hackers, and scammers, recent studies have shown that traditional digital literacy tools—guidelines like the CRAAP test—simply do not work. In actuality, the outdated tools are more detrimental than beneficial (Caulfield, 2016; Wineburg et al., 2020). It is risky to encourage the use of top-level domains, About pages, and update frequency as binary heuristics for authority. A site's beautiful appearance, absence of errors, and merriment of the presence of footnotes are all inadequate tactics for fact-checking, as are the notions that .orgs are inherently trustworthy (Valenza, 2020).

One distinctive aspect of the CCOW criteria lies in its comprehensive approach to source evaluation while the CRAAP test it focuses primarily on the content and structure of the source itself, rather than the broader context in which it was produced. For example, a source may meet all of the CRAAP criteria but still be biased or unreliable because it was funded by a special interest group with an agenda. In contrast, the CCOW criteria take into account the credentials, claims, objectives, and worldview of the author or publisher of the source, which can provide insight into potential biases or conflicts of interest. For example, we're researching a controversial medical treatment and come across a website that appears to provide helpful information. When applying for the CRAAP test, we might find that the website is current, relevant, authored by a reputable institution, accurate, and with a clear purpose. However, upon further investigation using the CCOW criteria, we discover that the website is funded by a pharmaceutical company that manufactures the treatment, and the information provided is biased in favor of promoting the treatment. This insight into the author's credentials and objectives can help us evaluate the source more critically and make a more informed decision about whether to trust the information provided.

One aspect to consider with the CRAAP test is its structured and formulaic nature, which may not always accommodate diverse sources or research inquiries. For example, a source may not fit neatly into the CRAAP categories of currency, relevance, authority, accuracy, and purpose, but may still be valuable for a particular research question. In contrast, the CCOW criteria are more flexible and adaptable to different types of sources and research questions. For example, when evaluating a blog post, the CCOW criteria could be used to assess the author's credentials, claims, objectives, and worldview, even if the post does not fit neatly into the CRAAP categories. While the CRAAP test can be a useful tool for evaluating sources, it has limitations in its narrow focus on content and structure, and its rigidity. The CCOW criteria offer a more comprehensive and adaptable framework for evaluating sources, taking into account both the content and context of the source.

The CRAAP Test has a flaw in that it assumes that the process of evaluating information is a one-way process, in which the evaluator is detached and unbiased, and can judge the information objectively by applying the CRAAP checklist. However, this approach overlooks the fact that the evaluator is not a neutral observer, but rather an active participant who brings their preconceptions, beliefs, and biases to the table. Therefore, their opinion of the information they are evaluating can be influenced by how well it aligns with their existing worldview. In other words, the evaluation of the information is a two-way interaction in which the evaluator and the information are in constant dialogue, and the evaluator's interpretation of the information is shaped by their background and experiences (Tardiff, 2022).

Review of literature

Nowadays, social media usage is increasing, and its diversity and mobility are contributing to this growth (Neilsen, 2012) it has emerged as the main information source and many use it for information acquisition and acts as an avenue for communication, enabling users to connect and share information in a pervasive and convenient way (Hong, 2015). According to the latest studies conducted by the Pew Research Center, young adults between the ages of 18 to 24, which is the

traditional age range for college students, are highly active in utilizing various social media platforms as part of their daily routines. The research indicates that this age group is particularly enthusiastic in adopting and frequently using a wide range of digital communication, entertainment, and information-sharing tools (Smith & Anderson, 2018; Auxier et al., 2022; Vogels et al., 2022) college students use social media as a source of news and information, and that social media can provide a means of accessing information that is not readily available through traditional media channels (Junco, 2012) similarly in a study conducted by Head et al. (2019); it was discovered that the majority of college students (85 %) obtained news from social media platforms, and 65 % did so from multiple platforms on a weekly basis although college students frequently turned to social media, as a source of breaking news and real-time information (Naaman et al., 2010), they are also commonly used by students to seek information for academic purposes (Manca & Ranieri, 2016), and to share those information with their peers (Kim & Kim, 2017). According to an article published in OpenLearn, social media provides three core benefits for academia: connection, communication, and collaboration (Chandler, 2021) by using social media platforms, students can connect with peers and experts in their fields of study, engage in meaningful conversations, and collaborate on academic tasks, learning from one another and developing important skills. A recent study by Pew Research Center found that YouTube is the most commonly used social media platform among teens, with 95 % of teens surveyed reporting the use of the platform, the study also found that other popular social media platforms include TikTok and Instagram, with 87 % and 71 % of teens using these platforms, respectively (Vogels et al., 2022). There are various other channels through which college students acquire information, Wikis, social networking sites, and media-sharing services, for instance; were discovered by Kim et al. (2011, 2014) to be common information sources. Research has indicated that Wikipedia has become a topic of interest across various social media platforms due to its popularity in academic research (Lim, 2009) it is frequently consulted as a reference tool for academic work (Head & Eisenberg, 2010) however recently there has been a notable shift in attention toward the potential of social question-and-answer platforms (social Q&A), such as Yahoo! Answers and Quora, to enhance collaborative information acquisition (Gazan, 2011).

In addition to considering the utilization and intention behind social media use, it is crucial to examine the approach employed to evaluate the information obtained from social media. The evaluation of information quality on social media presents additional hurdles due to the distinct features associated with these platforms. These challenges encompass the absence of traditional gatekeepers and mechanisms for quality control, the ambiguity surrounding the origin and context of information, and the convergence of diverse content types that can blur distinctions between sources and channels (Flanagin & Metzger, 2008) additionally the traditional cues that have been relied upon for evaluating the quality of information and its sources are frequently unclear or missing within the context of social media (Rieh, 2010; Wathen & Burkell, 2002). One significant worry is that certain students do not recognize the significance of information quality as a crucial aspect of their information-seeking behavior on social media (Kim & Sin, 2018) they often do not instinctively assess the credibility of the information they acquire (Braten et al., 2018), while they may initially choose information from sources that appear reliable, their evaluation standards may decrease if they lack access to more trustworthy sources (Kiili et al., 2008). The length of an answer and the inclusion of resources and personal scientific research experience greatly influence the evaluation of its quality. Moreover, the way emotions are expressed in an answer is a crucial criterion for assessing its overall quality when addressing mostly unanswered research topics, the inclusion of a conjecture or hypothesis becomes crucial in establishing the accuracy and legitimacy of the solution (Li et al., 2021), when evaluating youtube videos; Abed and Barzilai (2022) imply that students heavily depended on content evaluation and had little knowledge of trustworthy techniques for

evaluating the video's reliability, like sourcing and verification. Evidence indicates that credibility on Twitter is linked to several factors, including the authors of the tweets, the topic keywords, and the presence of retweets (Lin et al., 2016; Shariff et al., 2017). These studies underscore the significance of considering the reputation and expertise of tweet authors, the relevance of topic keywords, and the level of engagement indicated by retweets when assessing the credibility of information shared on Twitter. The studies imply that there is a lack of a uniform evaluation heuristic-based method in evaluating social media information that is devoid of a checklist approach since the checklist approach for evaluating information oversimplifies its complex nature and may overlook important nuances and context-specific factors as suggested by Metzger et al. (2010a, 2010b), this corroborates to Kim et al. (2021) who stated that these strategies are ineffective and inefficient and requires Information Literacy interventions.

Problem statement and research gap

While there is early evidence that students do take some evaluative actions when using information from social media (Head & Eisenberg, 2011; Kim & Sin, 2015); it is unclear whether students' evaluation strategies are effective in the eyes of library professionals and educator who have more experience and expertise in information literacy. Ignoring this platform in information literacy education could leave students ill-equipped to navigate the complexities of the digital information landscape effectively. Nonetheless, the current body of research on the evaluation of information acquired from social media platforms among college students is limited, indicating a gap in our understanding of how students efficiently and effectively evaluate the information they obtain from social media while the trustworthiness of information from such sources is often questionable and difficult to assess.

The current study is of considerable significance as it addresses the crucial issue of students' information evaluation strategies when using social media as a source of information. Moreover, the limited existing research on the evaluation of information acquired from social media platforms emphasizes the significance of this study in filling the knowledge gap. The research gap also encompasses the need for effective Information Literacy interventions to improve students' evaluation skills on social media. While some studies have pointed out the inefficiencies of checklist-based approaches, there is a need for context-specific strategies that empower students to critically evaluate social media information effectively; the lack of a uniform evaluation heuristic-based method in evaluating social media information suggests that existing approaches may not be sufficient in capturing the complex nature of information assessment on these platforms. Therefore, this study aims to address the research gap by employing the CCOW approach to the evaluation of social media information as it offers a comprehensive and nuanced analysis of how students can approach and assess social media information. By considering the alignment of the CCOW approach with established information literacy principles, the study seeks to identify areas for improvement and develop effective information literacy interventions that equip students with the necessary skills to effectively evaluate the complexities of social media information with discernment and accuracy. By using the CCOW approach, the study aims to equip students with critical evaluation skills tailored to the unique challenges posed by social media, ensuring they are better prepared to navigate the complexities of the social information landscape effectively.

Objectives

- To assess the current level of evaluation skills of college students regarding social media information.
- To examine the impact of the CCOW information literacy approach on the evaluation skills of college students regarding social media information.

Hypothesis

- There is a significant difference in the social media information evaluation skills of college students before and after the implementation of the CCOW information literacy approach.

Methodology

This study aims to investigate the impact of the CCOW (Credentials, Claims, Objectives, Worldview) approach to information literacy on the evaluation of social media information among college students. The present study employed a case study design, which allowed for an in-depth investigation of the impact of the CCOW intervention on the evaluation of social media information among college students. To facilitate the students' understanding of the CCOW approach, a specially designed CCOW Guide (<https://ccowguide.com/>) was utilized as a resource for the participants. It serves as a guide for students to acquire knowledge on the CCOW method for evaluating social media information. Before the intervention, a pre-test assignment on the Credentials of the author(s), Claims, Objectives, and Worldview of the social media information was given to the students without providing the CCOW Guide. This pre-test assignment was designed to assess the student's current level of evaluation skills concerning social media information. After the intervention, a post-test assignment was given to measure the impact of the intervention on the evaluation of social media information among the students.

A rubric was meticulously designed for this study and underwent a rigorous validation process. To ensure its reliability and credibility, the rubric was shared with subject experts in the form of an opinionnaire. This rigorous norming process helped establish a consensus among subject experts regarding the criteria and standards for evaluating the students' evaluation skills. By involving subject experts in rubric development and norming, the study ensured that the assessment tool effectively measured the targeted skills and provided reliable and valid results. It provides a framework for grading the assignments, ensuring consistency in the assessment process. At the end of the study, the effectiveness of the CCOW approach in improving students' evaluation skills was evaluated. The findings of the study will contribute to the development of better information literacy programs, specifically targeting social media information evaluation.

Process of intervention

To establish a baseline for the participants' evaluation skills, a pre-test assignment was administered prior to the intervention. This assignment focused on assessing the student's ability to evaluate the Credentials of the author(s), Claims, Objectives, and Worldview of social media information without access to the CCOW Guide. This allowed for a fair and accurate measurement of the student's initial proficiency in evaluating social media content. Following the pre-test, the intervention was conducted to introduce and implement the CCOW approach. During the intervention phase, a structured approach was implemented to assess the impact of the CCOW (Credentials, Claims, Objectives, Worldview) intervention on the evaluation of social media information among college students. To facilitate students' understanding of the CCOW approach, a specially designed CCOW Guide, available at <https://ccowguide.com>, was utilized as a comprehensive resource. The guide served as a user-friendly manual, which provides clear instructions and practical examples on effectively applying the CCOW method to evaluate social media information. Following the intervention, a post-test assignment was conducted to evaluate the effectiveness of the CCOW intervention. By comparing the results of the pre-test and post-test assignments, the impact of the intervention on the participants' evaluation skills could be assessed, providing valuable insights into the effectiveness of the CCOW approach.

Throughout the intervention, meticulous attention was given to

maintaining consistency in the delivery of the CCOW Guide and the assessment process. The intervention aimed to empower college students with the necessary knowledge and skills to critically analyze social media information, enabling them to make informed decisions and navigate the vast and variable landscape of social media with confidence.

Research site

Bormanik College was established in the year 1991 initially as Upper Shillong College. The name was changed to commemorate one of the freedom fighters during British Rule, the Late Bormanik Syiem. It is located in the state of Meghalaya, in the North Eastern part of India. The college is affiliated with North Eastern Hill University, Shillong, and it is equipped with >15 qualified staff, a college library, and an enrolment of >500 students.

Research sampling

The college under study offers a Higher Secondary and a Bachelor's degree program in Arts, adhering to a curriculum framework established by the Central University to which the college is affiliated. This affiliation ensures that the college follows the guidelines and standards set forth by the Central University, maintaining the quality and consistency of education across its programs. The curriculum has been meticulously designed to ensure that each student attends classes exclusively in their designated semester, with no overlap of students from different semesters. This strict structuring guarantee that students are precisely aligned with their appropriate courses and progress through the curriculum seamlessly, with no cross-semester class interactions. The college has a total enrollment of 500 plus students enrolled in different courses offered by the college, however, the current research employed a convenience sampling method to select participants for the study due to its practicality and accessibility, allowing for the inclusion of readily available participants within the specified course. The research subjects comprised a representative cohort of 120 students enrolled in the Bachelor of Arts program from the second (60 students) and fourth (60 students) semesters, encompassing all students from both selected semesters as research subjects for the study. Notably, students from the sixth semester were excluded from the study due to their status as outgoing students during the research period.

Data analysis approach

Predicated on the rubric scale, each criterion was assigned scores of 1 to 5, and then the measurement of central tendency was calculated to determine the mean and the standard deviation of the scores. The non-parametric Wilcoxon signed-rank test was conducted in R Studio to compare pre-test and post-test scores. This test is used to analyze related samples or repeated measurements from the same group. It also serves as a non-parametric alternative to the paired t-test and was introduced by Frank Wilcoxon in 1945. The test procedure involves collecting pre-test and post-test scores from the same participants. The differences between the scores are calculated and ranked based on their absolute values, with tied values receiving the average rank. Following the ranking of differences the Wilcoxon signed-rank test statistic is computed. It is then compared to a critical value at a chosen significance level (usually $p < .05$). If the test statistic is smaller than the critical value, it supports the alternative hypothesis or rejects the null, showing a significant difference between pre-test and post-test scores. This indicates that the intervention had a meaningful impact, leading to noticeable score changes. Hence, the Wilcoxon signed-rank test is a powerful tool for comparing pre-test and post-test scores within the same group. It helps us investigate whether there is a significant difference after an intervention or treatment, indicating its impact on the participants. In this study a critical P- value of <0.05 is considered to be statistically

significant.

Table 1 presents a clear picture of the demographic distribution of the 120 respondents. The majority of respondents were females (63.33 %) compared to males (36.67 %), the significant presence of female students (63.33 %) in the study can be attributed to the college's enrollment demographics, where a larger proportion of female students are enrolled compared to male students. As a result, the study sample represents the gender distribution within the college context, showcasing the prevailing trend of female predominance in the student body. Interestingly, the age group of 18–20 constituted the highest proportion of respondents (61 %), followed by the age group of 20–23 (34 %), and a small percentage of respondents below 18 years (2 %) and above 23 years (3 %). In terms of the semester, the respondents were equally distributed between the second and fourth semesters, with each group constituting 50 % of the total respondents.

The qualifications of the author or publisher, including their institutional affiliations, professional experience, or academic degrees, can influence the quality and accuracy of the information provided. The Credentials criteria of the CCOW method A credential is certified proof of an individual's qualifications or identity, establishing their credibility or authorization in a particular domain. In this study, it was found that the participants' scores on the Credentials criteria significantly improved from the pre-test (Mdn = 2) to the post-test (Mdn = 4) which indicates that the participants demonstrated a higher median score (Mdn) on the Credentials criteria after undergoing the CCOW intervention. The median score represents the middle score in a distribution of scores, and in this case, the post-test median score of 4 suggests an enhancement in the participants' ability to evaluate the credibility and expertise of the sources they encountered on social media. The improvement in scores was observed within the same set of participants, highlighting the effectiveness of the intervention in enhancing their evaluation skills.

A Wilcoxon signed-rank test revealed a statistically significant difference; $z = -10.95$, $p < .001$, $r = -0.71$. These results indicate that the observed disparity in median scores on the Credentials criteria between the pre-test and post-test assessments was highly unlikely to have occurred by chance. The Wilcoxon signed-rank test, a non-parametric test utilized to compare related groups, was employed to compare the median scores of the participants in the pre-test and post-test evaluations on the Credentials criteria. The substantial effect size indicated by a z -score of -10.95 suggests that the difference between the two assessments was not only statistically significant but also practically significant. Furthermore, the p -value of <0.001 indicates that the observed disparity between the two assessments reached statistical significance at a very high level of confidence, as displayed in Table 2.

The significance of examining the credentials of authors and information sources on social media has also been demonstrated by earlier studies. Gorodnichenko et al. (2021) asserted that determining the author's credentials and experience is essential for determining the legitimacy of content on social media. Similarly to this, Haim et al. (2018) discovered that one of the crucial standards for determining the validity

of material on social media is verifying the author's credentials.

The observed significant difference between the pre-test and post-test scores on the Credentials criteria emphasizes the effectiveness of the CCOW method in enhancing participants' ability to evaluate the credentials of information sources on social media. These compelling results offer strong evidence that the CCOW method was successful in improving participants' capacity to evaluate social media information based on the Credentials criteria.

The improvement in critical thinking skills is particularly relevant for the current era, where claims made through various media outlets can be difficult to evaluate. Evaluating claims of social media information is important because it enables individuals to determine the accuracy and truthfulness of the information presented. In the CCOW method, the Claims criteria refer to the factual accuracy of the information presented by the source. According to Tandoc et al. (2018), the ability to identify false claims is a critical component of media literacy in the digital age. Misinformation, disinformation, and fake news are prevalent on social media, and the ability to evaluate claims is essential in preventing the spread of false information.

Research has also shown that evaluating claims is a necessary step in the information evaluation process. In their study on the credibility of information on social media, Lee and Paek (2017) found that the accuracy of the information was the most critical factor influencing credibility. Similarly, a study by Metzger et al. (2010a, 2010b) revealed that individuals who evaluated the accuracy of claims were more likely to make informed decisions about the information presented.

The Wilcoxon signed-rank test was conducted to examine the difference in the Claims criteria scores among the participants themselves, comparing their Pre Test and Post Test scores. The results in Table 3 disclosed a significant difference, with the participants scoring significantly higher on the Claims criteria in the Post Test (Mdn = 5) compared to the Pre Test (Mdn = 1) ($z = -10.95$, $p < .001$), as visually represented in Fig. 2.

The effect size of the difference between the two groups was found to be large, with a correlation coefficient (r) of -0.71 . This indicates a strong relationship between the intervention and the improvement in critical thinking skills related to evaluating claims.

In the CCOW method, the Objectives criteria pertain to the purpose or motive behind the information presented. The findings of the study indicate that participants in the post-test condition had higher scores for the Objectives criteria (Mdn = 5) compared to the participants in the pre-test condition (Mdn = 2), as visually illustrated in Fig. 3. This suggests that after the intervention, the participants were better able to evaluate and identify the purpose behind the information presented on social media.

The ability to assess the objectives or motives of information is important in evaluating the credibility and reliability of social media information. According to Lee and Ma (2012), understanding the objectives of the sources can help individuals assess their biases and potential influence on the information presented. Similarly, Jan and Kim (2018) found that the purpose of the information affects individuals' perceptions of credibility and accuracy.

The Wilcoxon signed-rank test was used to compare the differences in the Objectives criteria scores between the pre-test and post-test assessments. The results, as indicated in Table 4, clearly illustrate that participants had significantly higher scores for the Objectives criteria in the post-test assessment compared to the pre-test assessment, with a median score of 5 in the post-test and 2 in the pre-test.

The statistical analysis shows that this difference was significant with a large effect size ($z = -10.95$, $p < .001$, $r = -0.71$). The negative value of the z -score indicates that participants in the post-test assessment had significantly higher scores than in the pre-test assessment. The p -value of <0.001 indicates that the observed difference in scores is highly unlikely to be due to chance, and the effect size of -0.71 suggests a large and meaningful difference between the two assessments.

The findings suggest that the data analysis using the Wilcoxon

Table 1
Demographic profile of participants.

Variable		Frequency N = 120
Gender	Male	36.67 %
	Female	63.33 %
Age group		100 %
	Below 18	2 %
	18–20	61 %
	20–23	34 %
	23 and above	3 %
	Total	100 %
Level of study	Second semester	50.00 %
	Fourth semester	50.00 %
	Total	100 %

Table 2
Credentials.

Outcome	Predictor	Group	Descriptive statistics			Test statistics			Effect size
			N	Mean	Standard deviation	W	z	p	r
Credentials	Factor	Pre Test	120	1.608	0.626	0	-10.954	< 0.001	-0.707
		Post Test	120	4.375	0.595				

Table 3
Claims.

Outcome	Predictor	Group	Descriptive statistics			Test statistics			Effect size
			n	Mean	Standard deviation	W	z	p	r
Claims	Factor	Pre Test	120	1.458	0.5	0	-10.954	< 0.001	-0.707
		Post Test	120	4.658	0.476				

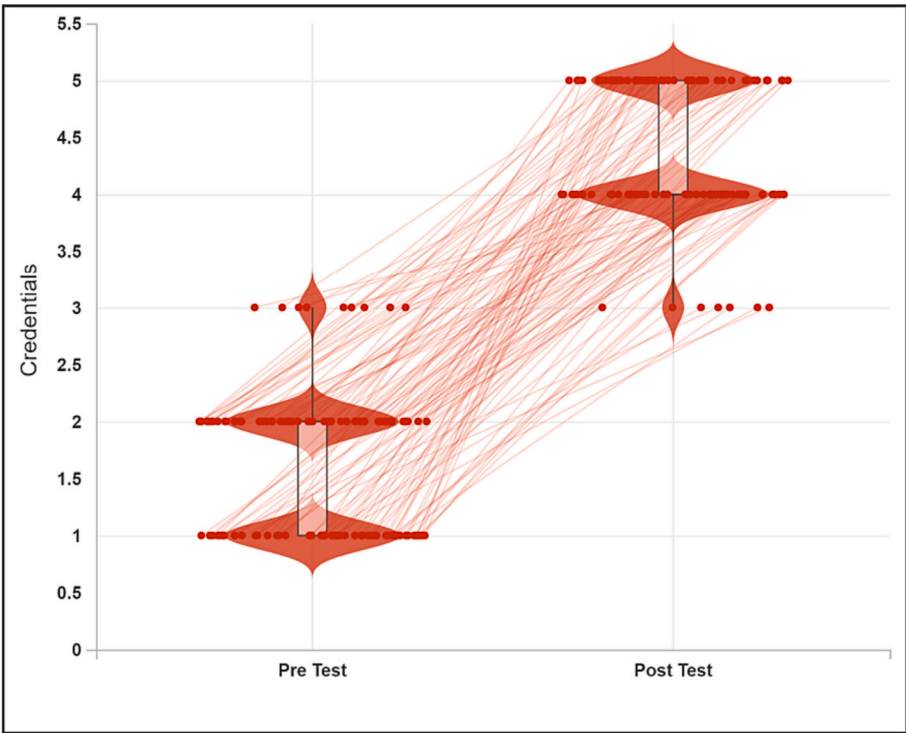


Fig. 1. Credentials.

signed-rank test shows a significant improvement in participants’ Objectives criteria scores from the pre-test to the post-test assessment, indicating the effectiveness of the CCOW method in enhancing their ability to assess the purpose or motive behind the information presented on social media.

The Worldview criteria in the CCOW method refer to the underlying beliefs, values, and assumptions that influence the information presented by sources. According to Tandoc et al. (2018), understanding the underlying beliefs and values of the sources can help individuals assess their potential biases and potential influence on the information presented. The test results showed a significant difference between the pre-test and post-test assessments, with a large effect size ($r = -0.7$). The median score for the Worldview criteria increased from 1 in the pre-test assessment to 5 in the post-test assessment, indicating that the intervention had a substantial impact on participants’ ability to recognize and evaluate the worldview or perspective presented in social media information, as visually depicted in Fig. 4.

The ability to identify the worldview or perspective presented in information is an important aspect of media literacy, as it can help

individuals understand the biases and potential influence on the information presented. This is particularly relevant in the context of social media, where information can be easily manipulated and presented from various perspectives. In summary, the data analysis shows that the CCOW method is an effective tool in improving individuals’ ability to identify the worldview or perspective presented in social media information. The Wilcoxon signed-rank test showed a statistically significant difference between the pre-test and post-test assessments for the Worldview criteria, as indicated in Table 5, with a z-score of -10.91 and a p-value of <0.001 ($p < .001$).

The significant difference found in the analysis suggests that the CCOW method was effective in improving the participants’ ability to evaluate the worldviews presented in social media information.

An analysis of the performance of students in the CCOW module showed that they were proficient in evaluating sources using the CCOW criteria. Additionally, the assessment of student work revealed various other noteworthy observations.

Based on the results of the CCOW module performance analysis, as illustrated in Fig. 5, it was found that the majority of students (51 %)

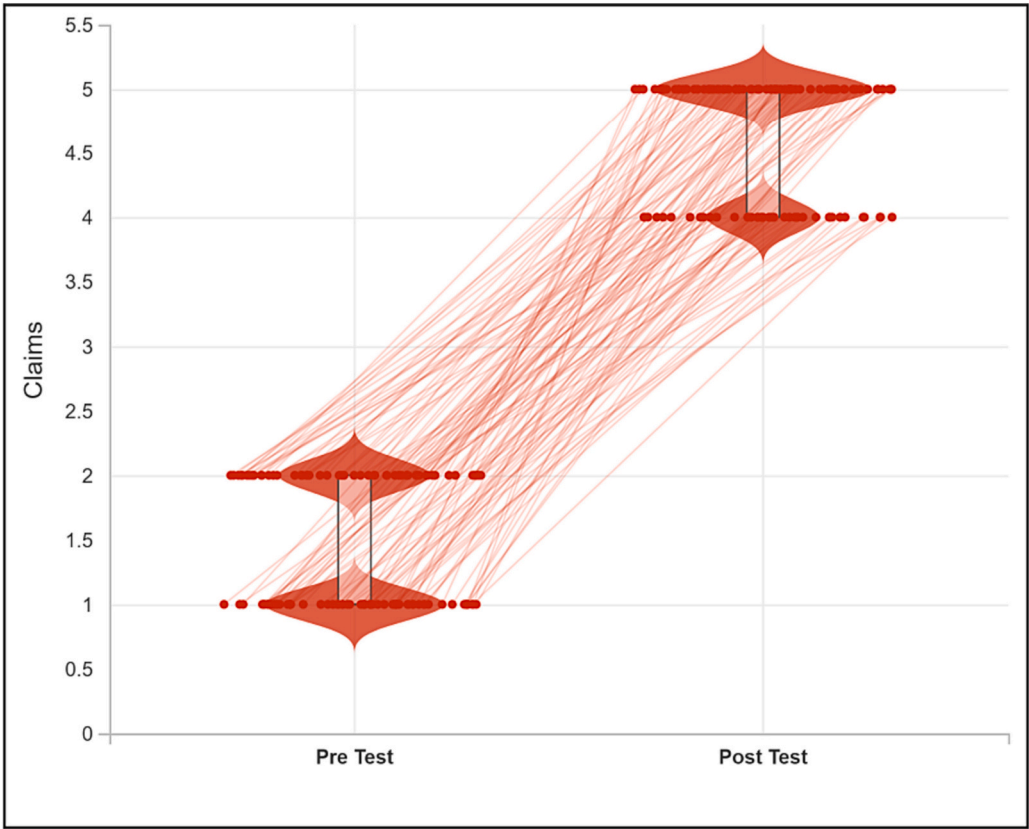


Fig. 2. Claims.

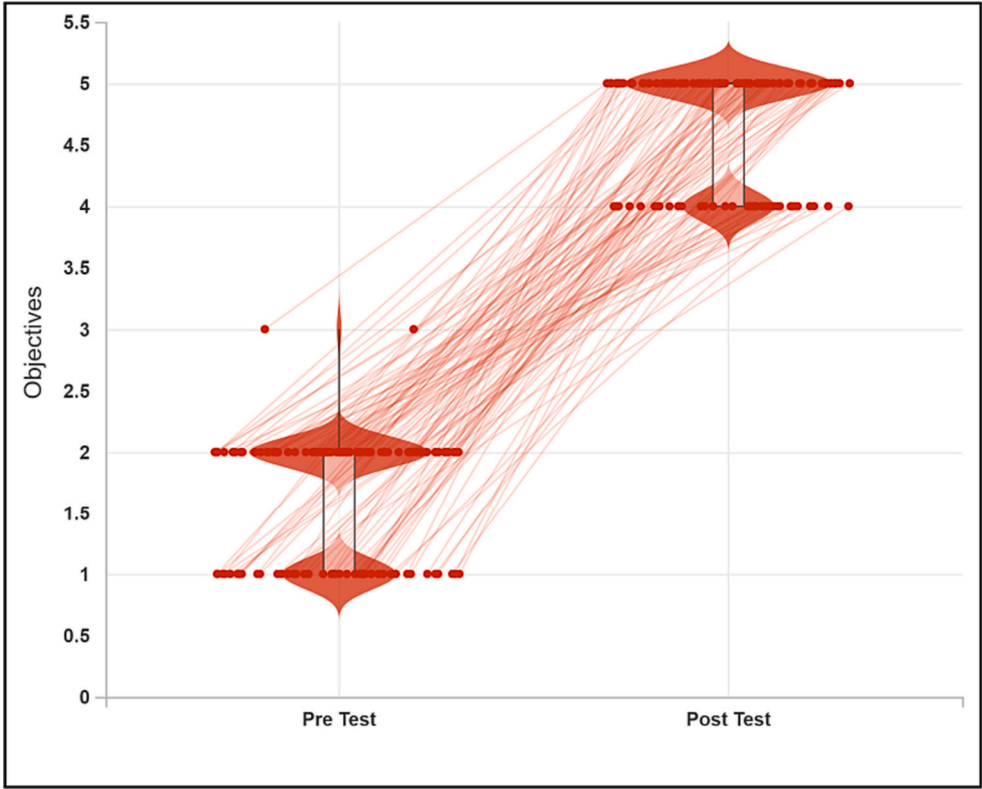


Fig. 3. Objectives.

Table 4
Objectives.

Outcome	Predictor	Group	Descriptive statistics			Test statistics			Effect size
			n	Mean	Standard deviation	W	z	p	r
Objectives	Factor	Pre Test	120	1.633	0.517	0	-10.954	<0.001	-0.707
		Post Test	120	4.692	0.464				

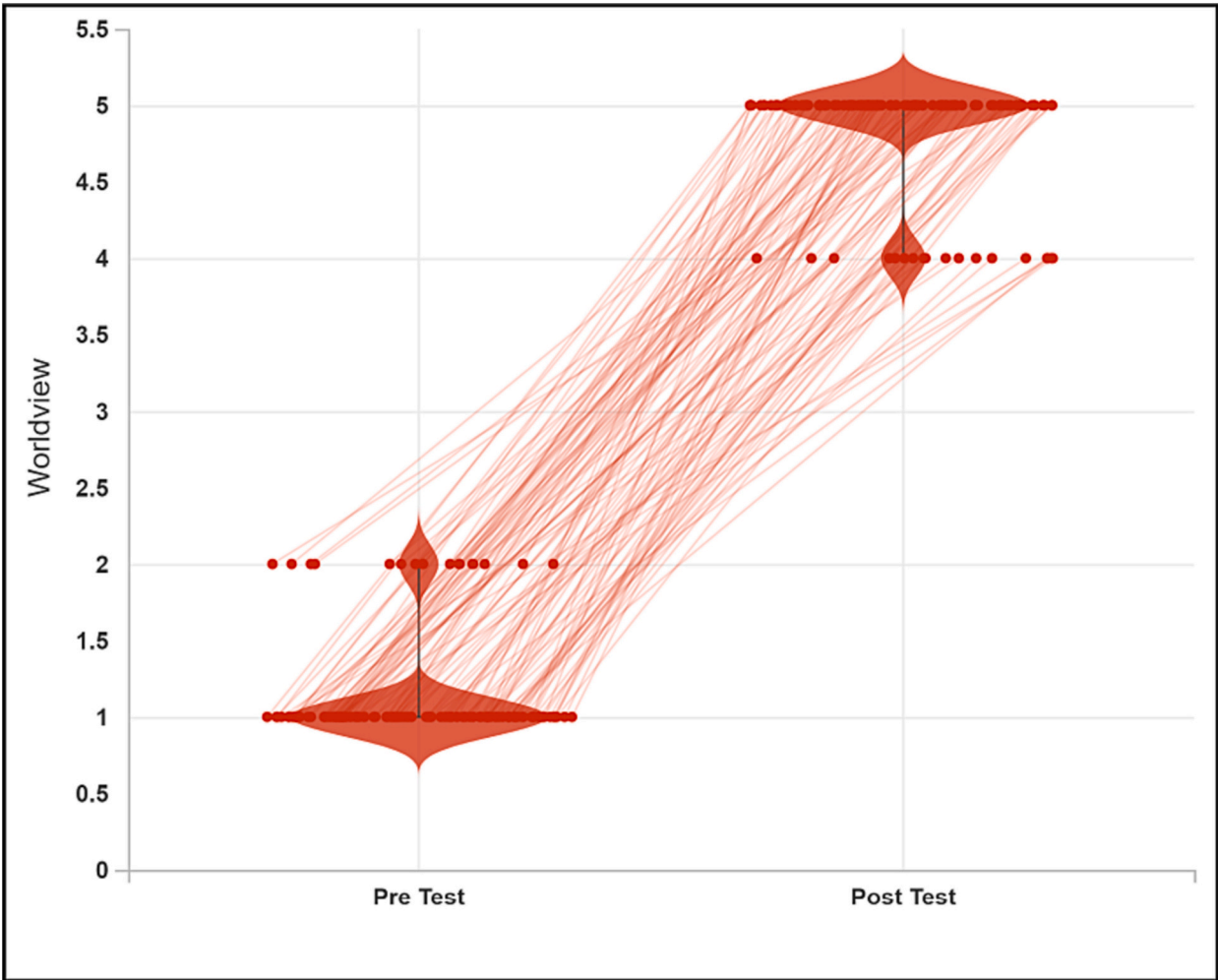


Fig. 4. Worldview.

Table 5
Worldview.

Outcome	Predictor	Group	Descriptive statistics			Test statistics			Effect size
			n	Mean	Standard deviation	W	z	p	r
Worldview	Factor	Pre Test	120	1.133	0.341	0	-10.91	<0.001	-0.704
		Post Test	120	4.85	0.359				

were able to meet the expectations set for the Credentials criterion. However, 6 % of the respondents were not able to meet the expectations, this is attributed to the reason that they mentioned some information about the author’s qualifications or background, but has not critically analyzed or assessed them in a meaningful way. On the other hand, a significant proportion of the respondents (43 %) exceeded the expectations for this criterion, demonstrating their advanced skills in evaluating sources based on credentials. While a majority of the students showed a readiness to investigate the author’s background beyond the

given source, and a strong ability to recognize aspects such as the author’s qualifications and experience to assess their credibility in discussing the subject matter, they also demonstrate the ability to assess how the author’s background and qualifications relate to the current discussion of the specific topic. This indicates that the students possess a comprehensive understanding of the author’s expertise, credibility, and the relevance of their work to the topic being discussed. Their ability to identify and evaluate the author’s qualifications and experience demonstrates a level of critical thinking that allows them to assess the value

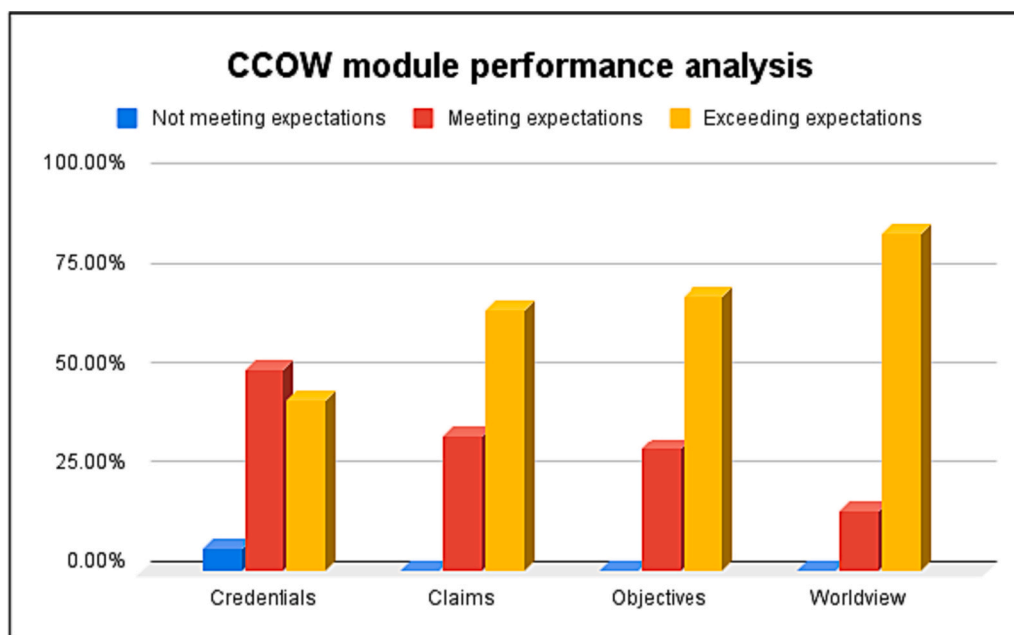


Fig. 5. CCOW module performance analysis.

of the author's contribution to the field. By doing so, they can draw well-informed conclusions and form opinions that are grounded in solid reasoning and evidence.

After the introduction of lateral reading, the majority of the students were able to demonstrate a comprehensive understanding of the content and evaluate the claims made in social media information. They displayed enthusiasm in leaving the source and consult other sources to verify the claims made in the social media information and evaluate its credibility. Specifically, 34.17 % of the students met the expected criteria, while a significant proportion of 65.83 % of the respondents exceeded the expectations, indicating a high level of competence in applying the lateral reading technique to evaluate the claim of social media information. The results of the CCOW module performance analysis indicate that the majority of students were able to successfully evaluate the Objectives criterion of social media information. Specifically, 30.83 % of students met expectations, while 69.17 % exceeded expectations in their ability to identify and evaluate the purpose or motive behind the information presented on social media using the CCOW method. It is important to highlight that a vast majority of students (85.00 %) demonstrated their ability to critically analyze social media information based on the worldview criterion of CCOW. Specifically, these students exhibited metacognition by engaging in discussions about both the source's worldview and their own. Furthermore, the students provided additional insightful analysis of how their worldview influenced their level of confidence or contempt in the source's information.

Discussion

The application of the CCOW approach in evaluating social media information yields significant insights into enhancing students' information evaluation skills. The study findings emphasize the relevance and impact of this approach, contributing to the student's ability to navigate the challenges posed by information on social media platforms. A particularly notable aspect illuminated by the study is the examination of the Credentials component within the CCOW framework, which surfaces as of paramount importance. The findings related to the Credential criterion align with broader research indicating that individuals often struggle to assess the credibility and authority of information sources (Freeman et al., 2020; Paul et al., 2017), particularly in

the context of social media. This challenge is exacerbated by the fact that the users often lack a clear understanding of the backgrounds and expertise of those disseminating information. This gap can lead to a deficiency in instinctively assessing the credibility of the acquired information, as noted by Bråten et al. (2017). However, through the application of the CCOW approach, there is a distinct shift toward placing substantial emphasis on information credibility. The approach equips students with the capability to discern reliable sources from those that are less trustworthy. This is evident from the compelling evidence presented in Fig. 1, showcasing a notable improvement in students' ability to critically evaluate the credibility of information sources, the research subjects move beyond traditional indicators like authorship and institutional affiliation, encouraging a broader perspective, urging the research subjects to dig deeper into the background, expertise, and potential biases of the information creators. Notably, social media's intrinsic nature encourages users to freely share claims and evidence for evaluating social media information, this characteristic grants social media a formidable role as an information hub, but it is often observed that as a story gains traction and circulates widely, its narrative tends to undergo distortions, consequently, the version of an event or research that reaches the users can be significantly divergent from the actual reality, presenting a substantially inaccurate portrayal (Brevard College Jones Library, 2023), it concurrently necessitates the cultivation of skills to verify the authenticity of sources, arguments, and presented evidence. The incorporation of the concept of lateral reading has yielded promising results, following its implementation; a significant majority of students demonstrated a comprehensive grasp of content and showcased proficiency in assessing claims found within social media information. This resonates with the research by Breakstone et al. (2021), affirming that lateral reading emerged as a superior approach in efficiently and accurately evaluating fact-checking claims within online information. The study's findings indicate that post-intervention, subjects expanded their inquiry beyond singular sources, engaging with multiple references to gauge expert consensus and verify claims. This innovative approach, known as lateral reading, complements various strategies while differing from others. For instance, it diverges from checklist methodologies such as the CRAAP test, which advocates meticulous analysis of original content (Meola, 2004; Brodsky et al., 2021). By incorporating lateral reading, students are encouraged to swiftly venture beyond initial sources, consulting various perspectives to fortify their information

evaluation process.

The study's insights into the realm of Objectives are equally compelling. In an era marked by the proliferation of sensationalized content and manipulative narratives on social media platforms, discerning the underlying intentions behind information dissemination becomes of paramount importance. The application of the CCOW approach effectively encourages students to not only identify potential biases and hidden agendas but also equips them with the tool to navigate content with a heightened sense of discernment. This alignment of the CCOW approach with the findings is particularly notable, revealing that the research subjects' evaluation skills exhibited a noteworthy enhancement. The CCOW framework prompts the research subjects to delve deeper into the purpose behind information creation and to move beyond merely responding to binary queries about the information's purpose. It encouraged the research subjects in active investigations to uncover the multifaceted motives behind the information's creation – whether it's meant to sell, amuse, deceive, or propagate a particular ideology. Furthermore, it's worth noting that the ACRL Framework for Information Literacy in Higher Education (2015) prominently underscores the role of metacognition in information literacy—defined as awareness and comprehension of one's own cognitive processes. While other evaluation methodologies have not given equal emphasis to this facet, the evident indications of its relevance (Tardiff, 2022) further reinforce the imperative to integrate metacognition into the evaluation of social media information. Within this framework, the concept of worldview emerges as a potent and approachable foundation, actively nurturing the growth of metacognition, this synergy highlights the paramount importance of metacognitive development. The study's outcomes highlight a notable change post-intervention, participants exhibited an increased awareness of factoring in both their own worldview and the perspectives of others when evaluating social media content. This transition from outright rejection to a more comprehensive evaluation aligns with Gray's (2011) assertion that in situations with conflicting worldviews, it's vital to recognize not just your own values but also those of others. This alignment underscores the significance of adopting a comprehensive approach when engaging with diverse viewpoints during the process of information evaluation.

Limitations

A limitation associated with the research is the restriction to students exclusively from the Bachelor of Arts program, despite the institution offering diverse educational levels. This conscious selection to constrain the scope might result in an incomplete grasp, inadvertently leaving out the wide range of perspectives and experiences of students from a myriad of academic disciplines and educational levels. As a consequence, this limited approach may not adequately capture the overall diversity and dynamics of the student community. Another limitation of the study lies in the necessity for comprehensive training of educators and librarians to effectively facilitate students in the evaluation of social media information. This imperative training ensures the consistent integration of the CCOW approach into the daily consumption of social media information among students. Thus, to augment the research's scope and depth, there is a need for future research to conduct subsequent inquiries that encompass a wider spectrum of academic domains and educational phases, additionally, it is crucial to spearhead research initiatives that focus on delivering comprehensive training on evaluation of social media information using the CCOW approach to educators and librarians alike.

Conclusion and recommendation

Information is crucial for various activities, but it is equally important to ensure that the information is valid, reliable, authoritative, and pertinent. Since anyone can publish anything online, the data gleaned from this source may be false or even deceptive. It's crucial to

understand how well pupils can evaluate information because, in most circumstances, teachers are powerless to regulate the caliber of Internet-derived information used by students (Kiili et al., 2008). When information goes through a process of filtering, reviewing, authentication, and evaluation, students are more likely to trust the source and accept an expert's assessment that it is valid and authoritative. However, in cases where the information has not undergone these processes, the students must take on the responsibility of assessing and evaluating the information themselves (Singh, 2004). But research has revealed that students frequently depend on flimsy and ineffective heuristics. For instance, rather than considering the message's content, they would assess a message's reliability only on its length (Kim & Sin, 2016; Lim, 2013; Lim & Simon, 2011). The study's findings provided insight into how college students, particularly those in the developing region of North Eastern India evaluate social media information. The findings suggest the need for further development of information literacy training. Information literacy programs must evolve and improve their approaches to assist students in efficiently utilizing and evaluating information from social media (Kim et al., 2021). Undoubtedly, both CCOW and the CRAAP test are valuable approaches for evaluating the credibility and reliability of sources. CCOW is a more comprehensive and nuanced approach that takes into account not only the traditional criteria of accuracy and reliability but also the source's objectives and worldview. The CCOW framework encourages students to consider not only what the source is saying but also why they are saying it and from what perspective, taking metacognition into utmost consideration. Asking oneself directly, "Why do I feel what I feel about this piece of information?" is essential to effectively evaluating information and avoiding typical biases. This self-examination is crucial for recognizing and overcoming biases like in-group prejudice and confirmation bias, which can impede correct information appraisal (Tardiff, 2022) this approach helps students to understand the context and biases that may influence the information presented and to evaluate it more critically. In contrast, the CRAAP test approach does not encourage students to engage in deeper critical thinking and may not be as effective in evaluating sources with more complex objectives or worldviews. Overall, CCOW provides a more comprehensive and nuanced approach to evaluating sources, which can be particularly beneficial in today's world of social media, where information is often presented in complex and nuanced ways.

As the amount of information available on social media continues to grow, the ability to evaluate sources and information becomes increasingly crucial in both academic and everyday contexts. Based on the literature and our findings, it is clear that evaluating social media information is an essential skill for students in the digital age. The use of CCOW as an information literacy intervention has shown promise in helping college students develop these skills. Our study demonstrated that the majority of students were able to successfully evaluate sources using the CCOW criteria, and a significant proportion exceeded expectations in all four criteria. Furthermore, students were able to exhibit metacognitive skills by considering both their worldview and that of the source, which influenced their level of trust or distrust in the information presented. While our study focused on college students, these findings may apply to other student populations. However, it is important to note that information literacy interventions may need to be tailored to specific populations and contexts. In addition, ongoing efforts are needed to promote the importance of evaluating social media information and to provide students with the necessary skills and resources to do so effectively. Overall, our study highlights the importance of incorporating information literacy interventions, such as the CCOW approach, into college curricula to promote critical thinking and digital literacy skills among students.

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